Arista Access Node ELEC-ACC-01 SOP: Resolving Hardware/Environmental Faults (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss) - Warning Severity

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_7696

Description

The Arista Access Node ELEC-ACC-01, situated at Electronic City Hub, is configured to operate in a redundant topology, ensuring high availability and reliability. However, in the event of a hardware/environmental fault, such as ASIC memory parity errors, line card crashes, or power supply redundancy loss, the node may experience a warning severity alarm (ALARM_7696). This is due to the node's reliance on ASIC-based architecture, where memory parity errors can occur, potentially affecting the node's control plane operations. Furthermore, the Arista EOS (Extensible Operating System) utilizes a distributed architecture, where line cards and power supplies operate in a redundant fashion. In the event of a failure, the node will attempt to reroute traffic and activate redundant components to minimize downtime. Nevertheless, a power supply redundancy loss can lead to a loss of redundancy, causing the node to operate in a non-redundant state, thus triggering the warning alarm. It is essential to promptly identify and resolve the root cause of the issue to prevent further complications and ensure seamless network operation.

Pre-requisite Checks

- Pre-Requisite: Verify redundant link status and confirm that all links are operational

- Pre-Requisite: Ensure remote hands are dispatched to Electronic City Hub for physical access and assistance, if required

- Pre-Requisite: Check power supply redundancy and confirm that backup power sources are available and functional

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes due to software or hardware issues	Medium
Power supply redundancy loss or failure	Low

Diagnostic Commands & Expected Outputs

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Command: show system resource, Expected Output: System resource
utilization and memory parity error information, Description: This
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command will provide information on system resource utilization, including memory parity errors, which can help identify the root cause of the issue.

Command: show interfaces, Expected Output: Interface status and link utilization information, Description: This command will provide information on interface status and link utilization, which can help identify any issues with the redundant links.

Command: show power, Expected Output: Power supply status and redundancy information, Description: This command will provide information on power supply status and redundancy, which can help identify any issues with the power supply redundancy.
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Resolution Steps

Step 1: Step: Replace failed ASIC or line card, Command: replace system asic, Description: This step will involve replacing the failed ASIC or line card with a new one, which will resolve the issue and prevent further complications.

Step 2: Step: Reboot the node, Command: reboot, Description: This step will involve rebooting the node to ensure that all changes take effect and the node is operating in a stable state.

Step 3: Step: Verify power supply redundancy, Command: show power, Description: This step will involve verifying that power supply redundancy is functioning correctly and that backup power sources are available and functional.

Post-Check Verification

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Command: show system resource, Expected Output: System resource utilization and memory parity error information, Description: This command will provide information on system resource utilization, including memory parity errors, which can help verify that the issue has been resolved.

Command: show interfaces, Expected Output: Interface status and link utilization information, Description: This command will provide information on interface status and link utilization, which can help verify that the redundant links are operational.

Command: show power, Expected Output: Power supply status and redundancy information, Description: This command will provide information on power supply status and redundancy, which can help verify that power supply redundancy is functioning correctly.
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Roll-back Procedures

- Step: Restore original ASIC or line card, Command: replace system asic revert, Description: This step will involve restoring the original ASIC or line card, which will revert the changes made during the resolution process.

- Step: Verify node status, Command: show system status, Description: This step will involve verifying that the node is operating in a stable state and that all changes have taken effect.